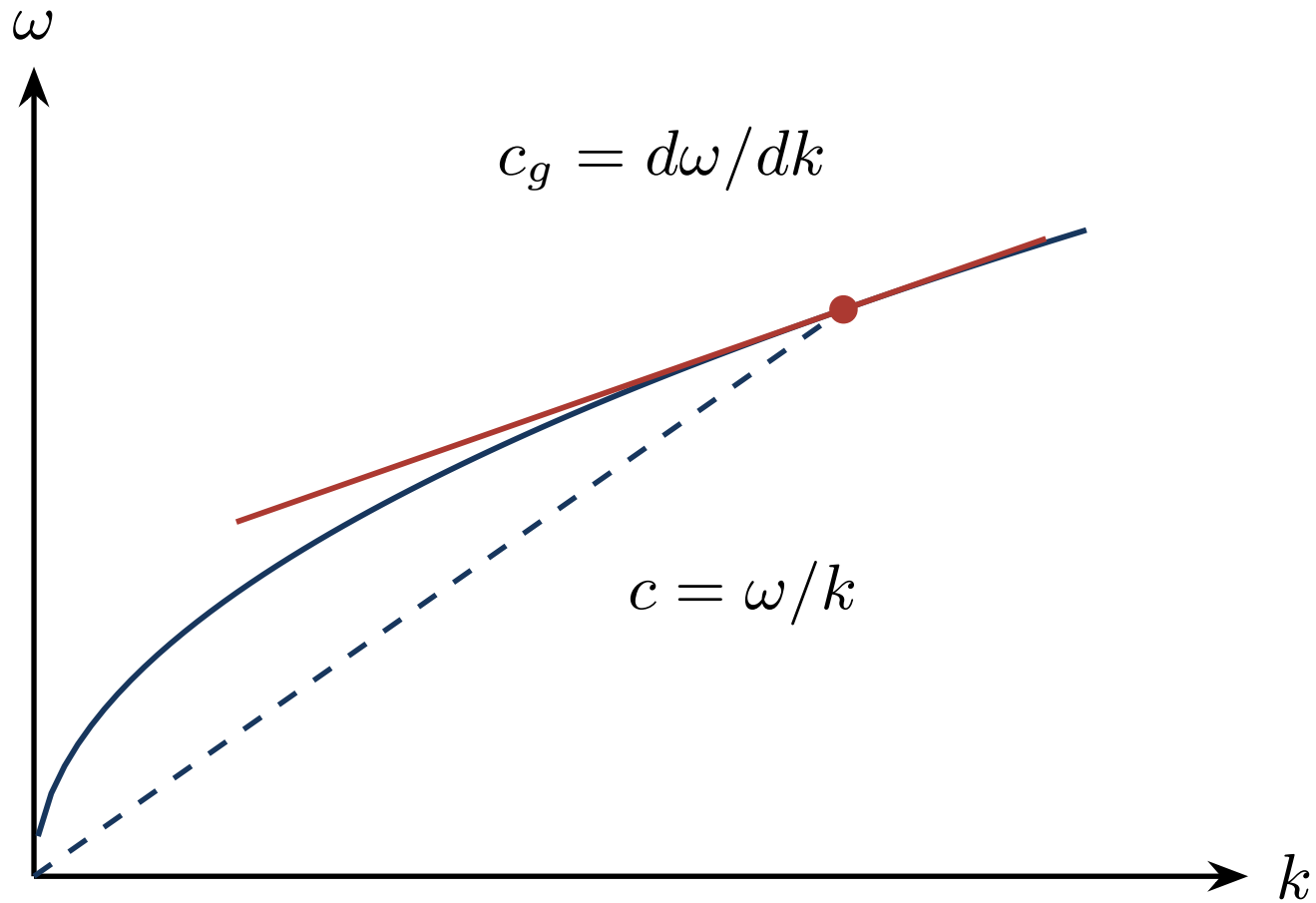

$\omega = \sqrt{gk}$；深水群速度 $c_g = c/2$